

MGMT 670, Sample Practice Final Exam

Fall 2025

Name (Print): _____

Student ID No.: _____

EXAM Rules:

This is an open-book, open-notes exam. Bring your own calculator.

No cellphone. No internet. No cheating!

PART I. MULTIPLE CHOICES

1. Circle your answers on the exam.

PART II. WORK PROBLEMS

1. Give neat, clean, and well organized answers.
2. Show your work to support your answers.

**This set of problems includes more questions than our actual final exam.*

** This practice exam focuses primarily on topics covered after the midterm, whereas the actual final exam will be comprehensive.*

Part I: Multiple Choice (35 Questions)

Identify the letter of the choice that **best** completes the statement or answers the question.

1. A machine that produces a major part for an airplane engine is monitored closely. In the past, 15% of the parts produced would be defective. The manager wants to determine the proportion of parts currently produced that are defective. What is the minimum sample size he should take so that at 95% confidence the margin of error will not be more than 0.08?
 - a. 77
 - b. 102
 - c. 139
 - d. 185
2. A department store is considering a new credit policy to try to reduce the number of customers defaulting on payments. A suggestion is made to discontinue credit to any customer who has been late with his/her payment twice. Past records show that 3% of all customers default. Amongst the defaults 95% were late at least twice. Also, 10% of those who have not defaulted have been late twice. The department store is considering to adopt this policy if 50% or more of the people with 2 late payments default. Given a person with 2 late payments, how likely is that he/she will default.
 - a. 0.0285
 - b. 0.0970
 - c. 0.1255
 - d. 0.2271
3. Regarding t-distribution, which one of the following is incorrect?
 - a. The mean is always positive
 - b. When the degrees of freedom become larger, the t-distribution approximates the normal distribution
 - c. t-distribution is symmetric
 - d. When the degrees of freedom $n_1 > n_2$, the value of $t_{\alpha/2, n_1}$ is always smaller than $t_{\alpha/2, n_2}$
4. For a public opinion poll for a close president election, let p denote the proportion of voters who favor candidate A. We wonder how large a sample should be taken if we want the maximum error of the estimate of p to be equal to
 - (i) 0.03 with 90% confidence,
 - (ii) 0.03 with 95% confidence, and
 - (iii) 0.02 with 95% confidence.

Suppose n_1 is the minimal sample size to achieve (i), n_2 is the minimal sample size to achieve (ii), and n_3 is the minimal sample size to achieve (iii), which of following is true?

- a. $n_1 \leq n_2 \leq n_3$
- b. $n_3 \leq n_2 \leq n_1$
- c. $n_2 \leq n_1 \leq n_3$
- d. $n_1 \leq n_3 \leq n_2$

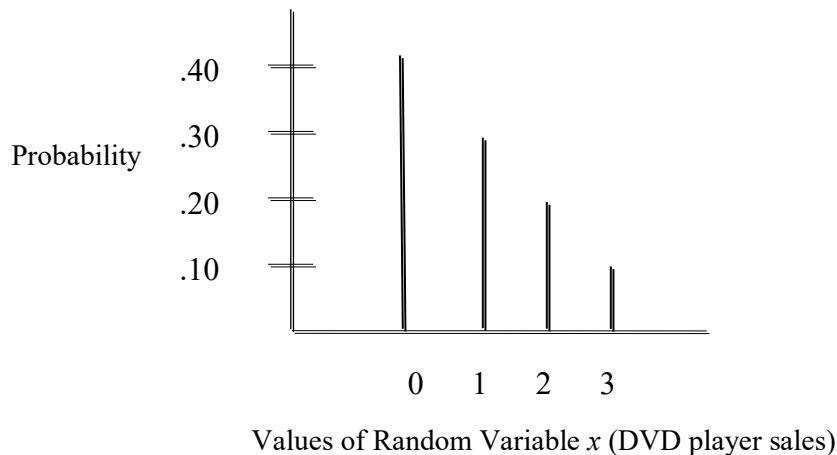
Answer the next two questions based on the following Exhibit

Exhibit

The freshman SAT scores in English at a reputed university in Indiana is normally distributed with a mean of 540 and a standard deviation of 15.

5. If nine SAT scores are selected at random from this university, the probability that the mean of this sample will be at least 550 is:
 - a. 0.9772
 - b. 0.0228
 - c. 0.4772
 - d. 0.7475
6. If one SAT score was selected at random (instead of nine), the approximate probability that it will be at least 550 is:
 - a. 0.9772
 - b. 0.0228
 - c. 0.7475
 - d. 0.2514
7. Neo wants to know the average age of residents in the greater Lafayette area. It is known that the standard deviation of people's ages is 7 years. What is the minimum sample size he should take so that at 90% confidence the margin of error will not be more than 1.2?
 - a. 92
 - b. 93
 - c. 9
 - d. 10
8. The manager of a travel agency is considering a new promotion plan in order to increase sales. Currently, the mean sales rate per agent is no more than 22 trips per month. The correct set of hypotheses for testing the effect of the promotion plan is
 - a. $H_0: \mu = 22$ $H_a: \mu \neq 22$
 - b. $H_0: \mu \geq 22$ $H_a: \mu < 22$
 - c. $H_0: \mu \leq 22$ $H_a: \mu > 22$
 - d. $H_0: \mu > 22$ $H_a: \mu \leq 22$
9. Suppose the mean of a population is 30. A sample of size 40 is taken and the sample mean is computed as 42. If $H_0: \mu \leq 34$, and you decided not to reject the null hypothesis, it means
 - a. you have committed a Type I error.
 - b. you have committed a Type II error.
 - c. you have committed either Type I or Type II error.
 - d. you have committed neither Type I nor Type II error.

10. Suppose we have a population that is normally distributed, and its standard deviation is unknown. We want to test the following hypotheses: $H_0: \mu=12$ vs $H_a: \mu \neq 12$. If the value of the test statistic is -1.73 based on a sample of size 20, what type of test statistic is it and what is the corresponding p-value?
- z statistic; 0.0287
 - z statistic; 0.0574
 - t statistic; approximately 0.10
 - t statistic; approximately 0.05
11. The diameter of 3.5 inch diskettes is assumed to be normally distributed. The quality control inspector conducted a hypothesis test to examine the average diameter of diskettes. He used a sample of size 35 and the following null and alternative hypotheses: $H_0: \mu = 3.5$ vs $H_a: \mu \neq 3.5$. At $\alpha=0.05$, he rejected the null hypothesis. Which one of the following statements is true if it is assumed that he/she performed the test correctly?
- The 95% confidence interval constructed by the sample did not contain 0
 - The 95% confidence interval constructed by the sample did not contain 3.5
 - The absolute value of the test statistic is less than 1.96
 - The p-value is greater than 0.05
12. The sales of DVD players at Venus music store were studied, and the probability distribution is shown in the following graphical representation. If X is a random variable denoting the number of DVD players sold in a week, then the mean number of DVD players sold in a week is:



- 0.6
 - 1.0
 - 1.5
 - 2.0
13. In testing $H_0: \mu_1 - \mu_2 = 0$ against $H_a: \mu_1 - \mu_2 \neq 0$, we obtained the value of the z-test statistic to be 1.46. Given that $\text{Prob}(z > 1.46) = 0.072$. Then which of the following conclusion is correct?
- Reject H_0 at $\alpha = 10\%$ since p-value = 0.072 < 0.10
 - Cannot reject H_0 at $\alpha = 5\%$ since p-value = 0.072 > 0.05
 - Cannot reject H_0 at $\alpha = 10\%$ since p-value = 0.144 > 0.10
 - None of the above is correct.

14. A law enforcement agent claims that at least 88% of the drivers stopped on Saturday nights for speeding are under the influence of alcohol. A sample of 66 drivers who were stopped for speeding on a Saturday night was taken. Eighty percent of the drivers in the sample were under the influence of alcohol. What is the value of the test statistic?
- a. 2.0
 - b. -2.0
 - c. 1.89
 - d. 1.78

Answer the next three questions based on the following exhibit

Exhibit

The IT director of a large company is concerned about the implication of computer idle time. According to the company data, the mean idle time of the computers in the company is 68 minutes. The system administrator examines the records of 42 computers randomly selected from the purchasing group and finds that their mean idle time is 71.2 minutes with a sample standard deviation of 18 minutes. The IT director wants to determine whether the mean idle time of the computers in the purchasing group is higher than the company average.

15. What is the value of the test statistic?
- a. 0.125
 - b. 1.152
 - c. 2.000
 - d. 2.512
16. What is the conclusion at 10% level of significance?
- a. There is not enough evidence to conclude that the mean idle time for computers in the purchasing group is higher than the company average.
 - b. There is enough evidence to conclude that the mean idle time for computers in the purchasing group is higher than the company average.
 - c. No conclusion can be made with the given information.
 - d. None of the above is true.
17. What should be the minimum average idle time of a sample of 42 computers for you to reject the null hypothesis at 10% level of significance (assuming that the sample standard deviation is 18 minutes)?
- a. 64.445
 - b. 68.000
 - c. 71.618
 - d. None of the above.

Answer the next three questions based on the following exhibit

Exhibit

Two major automobile manufacturers A and B have produced compact cars with the same size engines. We are interested in determining whether there is a significant difference in the MPG (miles per gallon) of the two brands of automobiles. A random sample of 4 cars from each manufacturer is selected, and 4 drivers (Dave, Jason, Rick and Sam) are selected to drive each manufacturer's automobile for a specified distance. Assume that the differences normally distributed. The following data show the results of the test.

Manufacturer A	Manufacturer B
32	34
38	35
36	36
34	31

Define the following:

μ_1 is the mean of the MPG of automobiles of Manufacturer A,

μ_2 is the mean of the MPG of automobiles of Manufacturer B,

μ_d is the mean of the differences of the MPG between automobiles of Manufacturer A and B.

18. What are the appropriate null and alternative hypotheses?
 - a. Independent samples, $H_0: \mu_1 - \mu_2 = 0$, $H_a: \mu_1 - \mu_2 \neq 0$
 - b. Paired samples, $H_0: \mu_d = 0$, $H_a: \mu_d \neq 0$
 - c. Independent samples, $H_0: \mu_1 - \mu_2 \leq 0$, $H_a: \mu_1 - \mu_2 > 0$
 - d. Paired samples, $H_0: \mu_d \leq 0$, $H_a: \mu_d > 0$
19. What is the value of the test statistic?
 - a. 2.832
 - b. 2.450
 - c. 1.633
 - d. 0.816
20. At 10% level of significance, the conclusion of the test is that
 - a. there is no significant difference between the MPGs of the two brands of automobiles.
 - b. there is significant difference between the MPGs of the two brands of automobiles.
 - c. we reject the null hypothesis.
 - d. none of the above.
21. In a simple regression analysis (where Y is a dependent and X an independent variable), if the Y intercept is positive, then
 - a. there is a positive correlation between X and Y.
 - b. if X is increased, Y must also increase.
 - c. if Y is increased, X must also increase.
 - d. none of the above alternatives is correct.

22. Larger values of r^2 imply that the observations are more closely grouped about the
- average value of the independent variables
 - average value of the dependent variable
 - least squares regression line
 - origin
23. If a 98% confidence interval for β_1 is constructed as $[1.75, 3.92]$ in a simple linear regression model, we conclude that
- At $\alpha=0.02$, there exists a negative linear relationship between the variables
 - At $\alpha=0.20$, there is no evidence of a significant linear relationship between the variables
 - At $\alpha=0.02$, there exists a significant linear relationship between the variables
 - None of the above

Answer the next four questions based on the following exhibit

Exhibit

The following data show the overall vacancy rates (x , %) and the average rental rates (Y , per square foot) for the central business district for 6 selected markets.

Market	x	Y	$(x-\bar{x})*(Y-\bar{Y})$	$(x-\bar{x})^2$	$(Y-\bar{Y})^2$
Atlanta	22	19	-77.92	84.03	72.25
Boston	6	34	-44.42	46.69	42.25
Chicago	16	24	-11.08	10.03	12.25
New York	11	37	-17.42	3.36	90.25
Philadelphia	15	19	-18.42	4.69	72.25
San Francisco	7	32	-26.25	34.03	20.25
Sum	77	165	-195.50	182.83	309.5

We obtain that $b_1 = \frac{SS_{XY}}{SS_X} = \frac{-195.50}{182.83} = -1.069$, $b_0 = \bar{y} - b_1\bar{x} = \frac{165}{6} - (-1.069)(\frac{77}{6}) = 41.22$.

24. The estimated regression equation that could be used to predict the average rental rate given the overall vacancy rate is
- Average Rental Rate = $41.22 - 1.069 \times \text{Vacancy Rate}$
 - Average Rental Rate = $41.22 + 1.069 \times \text{Vacancy Rate}$
 - Average Rental Rate = $-41.22 - 1.069 \times \text{Vacancy Rate}$
 - Average Rental Rate = $-41.22 + 1.069 \times \text{Vacancy Rate}$
25. Given that $SSR=208.94$, the F test statistic and critical F value at $\alpha = 0.05$ are
- 8.31 and 3.85
 - 8.31 and 7.71
 - 7.93 and 7.71
 - 7.93 and 7.71

26. The conclusion at the 0.05 level of significance is
- There is no significant relationship between vacancy rate and average rental rate
 - There is a cause and effect relationship between vacancy rate and average rental rate
 - There is a significant linear relationship between vacancy rate and average rental rate
 - None of the above
27. The overall vacancy rate in the central business district in Detroit is 20%. The expected rental rate for Detroit is
- 18.94
 - 11.84
 - 13.84
 - 19.84

Answer the next three questions based on the following exhibit

Exhibit

The starting salaries of recent graduates at a large university (LU) are normally distributed with a mean of \$30,000 and a standard deviation of \$8,000.

28. The minimum salary of the top 15% of the LU graduates is:
- 22,000
 - 38,320
 - 38,000
 - None of the above
29. Individuals with starting salaries of less than \$19,760 receive a low income tax break. What percentage of the LU graduates will receive the tax break?
- 0.50
 - 0.35
 - 0.20
 - 0.10
30. A sample of 15 graduates of LU is selected at random. What is the approximate probability that less than 3 graduates will receive the tax break?
- 0.816
 - 0.702
 - 0.625
 - None of the above.

31. Comparing the coefficient of multiple determination (R^2) and adjusted coefficient of multiple determination (R_a^2) for the same regression model, which of the following statements is correct?
- When an additional independent variable is added to the model, R^2 will always decrease.
 - When an additional independent variable is added to the model, R_a^2 will always increase.
 - R_a^2 can never be greater than R^2 .
 - Both (b) and (c) are right.
32. If we add a new independent variable to a multiple regression model, then
- other variables in the model will become significant if the new variable has a strong correlation with the dependent variable
 - the adjusted R^2 will increase
 - the variables remaining in the model will become less significant because more variables are included in the model
 - the R^2 will not decrease

Use the following exhibit to answer the next three questions

Exhibit

The manager of the human resources department collected the following data.

ID	Age	Salary (\$1,000)	Years of Experience	Job Satisfaction
1	23	45.15	21	17.34
2	31	38.94	10	14.16
3	64	34.06	25	30.31
4	46	69.87	21	23.22
5	34	54.15	24	18.09
6	39	88.28	12	24.59
7	31	61.34	24	14.47
8	19	55.92	3	11.03
9	33	63.71	18	15.99
10	26	34.29	26	18.95
11	44	50.63	18	17.77
12	18	62.18	10	18.19
13	21	36.89	17	9.29
14	34	70.20	9	14.28
15	45	50.00	22	25.16
16	23	73.36	7	18.04
17	25	28.87	13	9.04
18	44	31.94	11	14.81
19	52	49.50	23	20.27

In order to study the relationship between a worker's satisfaction with his or her age, salary, and experience, the following Minitab outputs were obtained.

Regression Analysis: Satisfaction versus Age, Salary, Experience

The regression equation is

Satisfaction = - 1.55 + 0.269 Age + 0.121 Salary + 0.219 Experience

Predictor	Coef	SE Coef	T	P
Constant	-1.549	4.183	-0.37	0.716
Age	0.26853	0.07617	3.53	0.003
Salary	0.12076	0.05302	2.28	0.038
Experience	0.2187	0.1408	1.55	0.141

S = 3.52842 R-Sq = 65.3% R-Sq(adj) = ☺☺☺

Analysis of Variance

Source	DF	SS	MS	F	P
Regression	3	350.73	116.91	9.39	0.001
Residual Error	15	186.75	12.45		
Total	18	537.48			

Source	DF	Seq SS
Age	1	274.22
Salary	1	☺☺☺
Experience	1	☺☺☺

New

Obs	Fit	SE Fit	95% CI	95% PI
1	12.688	1.978	(8.472, 16.904)	(4.066, 21.310)

Values of Predictors for New Observations

New

Obs	Age	Salary	Years
1	32.0	34.06	7.00

Regression Analysis: Age versus Experience, Satisfaction

The regression equation is

Age = 3.85 + 0.312 Experience + 1.44 Satisfaction

Predictor	Coef	SE Coef	T	P
Constant	3.849	7.430	0.52	0.611
Experience	0.3121	0.3446	0.91	0.379
Satisfaction	1.4354	0.4381	3.28	0.005

S = 8.97822 R-Sq = 53.4% R-Sq(adj) = 47.6%

Regression Analysis: Salary versus Experience, Satisfaction

The regression equation is

Salary = 46.8 - 1.15 Experience + 1.41 Satisfaction

Predictor	Coef	SE Coef	T	P
Constant	46.81	12.44	3.76	0.002
Experience	-1.1506	0.5771	-1.99	0.063
Satisfaction	1.4063	0.7337	1.92	0.073

S = 15.0340 R-Sq = 24.6% R-Sq(adj) = 15.2%

Regression Analysis: Age versus Salary, Experience

The regression equation is
 $\text{Age} = 17.6 + 0.045 \text{ Salary} + 0.869 \text{ Experience}$

Predictor	Coef	SE Coef	T	P
Constant	17.60	13.01	1.35	0.195
Salary	0.0448	0.1737	0.26	0.800
Experience	0.8685	0.4080	2.13	0.049

$S = 11.5813$ $R\text{-Sq} = 22.5\%$ $R\text{-Sq}(\text{adj}) = 12.8\%$

33. The value of the adjusted R^2 is
- 21.8%
 - 65.3%
 - 58.3%
 - 391.5%
 - 51.0%
34. The VIF associated with Age is
- 3.50
 - 1.15
 - 2.19
 - 1.29
 - 1.05
35. With a confidence level of 95%, which of the following intervals will contain the average satisfaction of employees with an Age of 32, Salary of 34,060 and an experience of 7 years?
- (8.472, 16.904)
 - (4.066, 21.310)

Part II: Work Problems (3 Problems)

Problem 1

A new computer software package has been developed. To evaluate the benefits of the new software package, 12 systems analysts are randomly selected. 6 of them are instructed to produce an information system by using the current technology. The other 6 analysts are trained in the use of the new software package and then instructed to use it to produce the information system. The completion time data (in hours) for the 12 analysts are shown below. The researcher-in-charge of the new software evaluation project wants to determine whether there is a significant difference between the mean project completion times of the current technology and the new software package. Assume that the project completion times of the two methods are independently normally distributed and have equal variances.

Current Technology: 30 28 35 38 37 36

New Software: 26 22 31 33 20 30

The following is the Minitab output:

Two-Sample T-Test and CI: Current Technology, New Software

Two-sample T for Current Technology vs New Software

	N	Mean	StDev	SE Mean
Current Technology	6	34.00	4.05	1.7
New Software	6	27.00	5.22	2.1

Difference = mu (Current Technology) - mu (New Software)

Estimate for difference: 7.00

95% CI for difference: (0.99, 13.01)

T-Test of difference = 0 (vs not =): T-Value = 2.60 P-Value = 0.027 DF = 10

Both use Pooled StDev = 4.6690

- What is the point estimate of the difference between the mean project completion times of the current technology and the new software package?
- State the appropriate null and alternative hypotheses for the researcher-in-charge.
- What is the value of the test statistic and what is your conclusion at $\alpha = 10\%$?
- What does the 95% confidence interval in the report for the difference between the mean project completion times of the current technology and the new software package, tell us?
- Use your answer in part (d) to make a conclusion for the hypothesis test.

Problem 2

The general manager of a chain of furniture stores believes that experience is the most important factor in determining the level of success of a salesperson. To examine this belief she records last month's sales (in \$1,000s) and the years of experience of 12 randomly selected salespeople. The data are listed below.

Salesperson	Years of Experience	Sales
1	7	13
2	8	8
⋮	⋮	⋮
11	8	11
12	17	18

Descriptive Statistics: Years of Experience, Sales

Variable	N	Mean	SE Mean	StDev	Minimum	Q1	Median	Q3	Maximum
Years of Experience	12	9.25	1.96	6.80	0.00	4.00	8.00	16.50	19.00
Sales	12	12.92	1.52	5.26	6.00	8.25	12.00	17.75	23.00

Regression Analysis: Sales versus Years of Experience

Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Regression	1	270.084	270.084	59.25	0.000
Years of Experience	1	270.084	270.084	59.25	0.000
Error	10	45.583	4.558		
Lack-of-Fit	5	9.916	1.983	0.28	0.907
Pure Error	5	35.667	7.133		
Total	11	315.667			

Model Summary

S	R-sq	R-sq(adj)	R-sq(pred)
2.13501	85.56%	84.12%	78.35%

Coefficients

Term	Coef	SE Coef	T-Value	P-Value	VIF
Constant	6.09	1.07	5.69	0.000	
Years of Experience	0.7290	0.0947	7.70	0.000	1.00

Regression Equation

Sales = 6.09 + 0.7290 Years of Experience

Prediction for Sales

Regression Equation

Sales = 6.09 + 0.7290 Years of Experience

Variable Setting
Years of Experience 11

Fit	SE Fit	95% CI	95% PI
14.1090	0.638217	(12.6870, 15.5311)	(9.14394, 19.0741)

Variable Setting
Years of Experience 12

Fit	SE Fit	95% CI	95% PI
14.8380	0.669088	(13.3472, 16.3288)	(9.85278, 19.8232)

Variable Setting
Years of Experience 13

Fit	SE Fit	95% CI	95% PI
15.5670	0.711319	(13.9821, 17.1519)	(10.5528, 20.5811)

Variable Setting
Years of Experience 14

Fit	SE Fit	95% CI	95% PI
16.2960	0.763025	(14.5958, 17.9961)	(11.2442, 21.3477)

Variable Setting
Years of Experience 15

Fit	SE Fit	95% CI	95% PI
17.0249	0.822421	(15.1925, 18.8574)	(11.9271, 22.1228)

- Determine whether there is a significant regression relationship between a salesperson's monthly sales and his/her years of experience. Use $\alpha = 1\%$.
- Consider the salesperson 11 (Years = 8, Sales = 11). Calculate predicted Sales and residual associated with the person.
- What's 95% interval estimate for the monthly sales of a **particular** salesperson with 14 years of experience?
- Would you reject the null hypothesis that $\beta_1 = 0$ (coefficient of Years of Experience is 0) at 5% significance level?
- Calculate 95% confidence interval for the coefficient of "Years of Experience" in the regression model. Can we reject the Null hypothesis that this coefficient is 0.45 with 5% level of significance?

Problem 3

A company obtained the data for the following variables for 93 employees:

Salary = beginning annual salary

Educat = years of schooling at the time of hire

Exper = number of months of previous work

Months = number of months after January 1, 1987 that the individual was hired

Gender = 1 for males and 0 for females

Mary, the HR director of the company, obtained results using several different regression models.

Regression Analysis: SALARY versus EDUCAT, EXPER, ...

The regression equation is

$$\text{SALARY} = 50391 + 994 \text{ EDUCAT} + 18.5 \text{ EXPER} + 298 \text{ MONTHS} - 3174 \text{ Gender} + 978 \text{ Edu*Gender}$$

Predictor	Coef	SE Coef	T	P
Constant	50391	4959	10.16	0.000
EDUCAT	994.4	384.2	2.59	0.011
EXPER	18.453	8.096	2.28	0.025
MONTHS	298.50	70.57	4.23	0.000
Gender	-3174	10395	-0.31	0.761
Edu*Gender	978.2	-----	☺☺	0.213

S = 6860.19 R-Sq = 51.4% R-Sq(adj) = 48.6%

Analysis of Variance

Source	DF	SS	MS	F	P
Regression	5	4330337692	866067538	18.40	0.000
Residual Error	87	4094416272	47062256		
Total	92	8424753964			

Regression Analysis: SALARY versus EDUCAT, EXPER, MONTHS

The regression equation is

$$\text{SALARY} = 42938 + 1900 \text{ EDUCAT} + 19.4 \text{ EXPER} + 269 \text{ MONTHS}$$

Predictor	Coef	SE Coef	T	P
Constant	42938	5187	8.28	0.000
EDUCAT	1899.6	374.9	5.07	0.000
EXPER	19.351	9.428	2.05	0.043
MONTHS	268.79	83.26	3.23	0.002

S = 8154.09 R-Sq = 29.8% R-Sq(adj) = 27.4%

Analysis of Variance

Source	DF	SS	MS	F	P
Regression	3	2507213192	835737731	12.57	0.000
Residual Error	89	5917540772	66489222		
Total	92	8424753964			

Regression Analysis: SALARY versus EDUCAT, EXPER, MONTHS, Gender

The regression equation is

$$\text{SALARY} = 47598 + 1233 \text{ EDUCAT} + 16.5 \text{ EXPER} + 306 \text{ MONTHS} + 9711 \text{ Gender}$$

Predictor	Coef	SE Coef	T	P
Constant	47598	4445	10.71	0.000
EDUCAT	1233.0	334.9	3.68	0.000
EXPER	16.506	7.972	2.07	0.041
MONTHS	306.07	70.54	4.34	0.000
Gender	9711	1598	6.08	0.000

S = 6882.50 R-Sq = 50.5% R-Sq(adj) = 48.3%

Analysis of Variance

Source	DF	SS	MS	F	P
Regression	4	4256300301	1064075075	22.46	0.000
Residual Error	88	4168453663	47368792		
Total	92	8424753964			

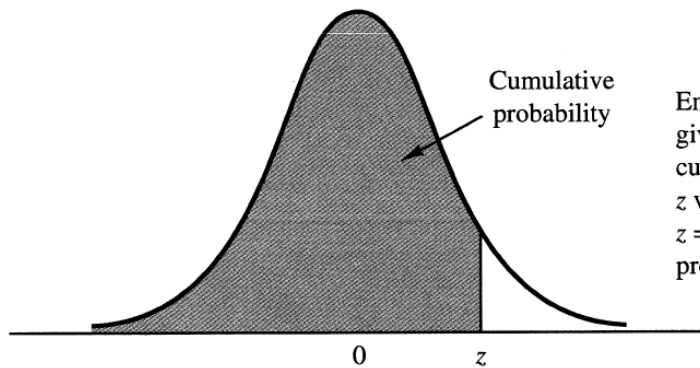
- a. Suppose the following model proposed by Mary is generally acceptable by the experts in the area:

$$\text{Salary} = \beta_0 + \beta_1 \text{Educate} + \beta_2 \text{Exper} + \beta_3 \text{Months} + \beta_4 \text{Gender} + \beta_5 \text{Educate*Gender} + \varepsilon$$

Conduct the F test for the overall fit of regression. State the hypotheses to be tested, the decision rule, critical number, and your decision. Use $\alpha = 5\%$.

- b. Is there a difference in salaries, on average, for male and female workers after accounting for the effects of the three other explanatory variables? State the hypotheses to be tested, the decision rule, and your decision. Use $\alpha = 5\%$.
- c. Assume that the first model on the previous page is adequate. What salary would you predict, on average, for males with 12 years of education, 10 years of experience, and with time hired equal to 15 months? What salary would you predict, on average, for females if all other factors are equal (i.e., 12 years of education, 10 years of experience, and 15 months of hiring)?

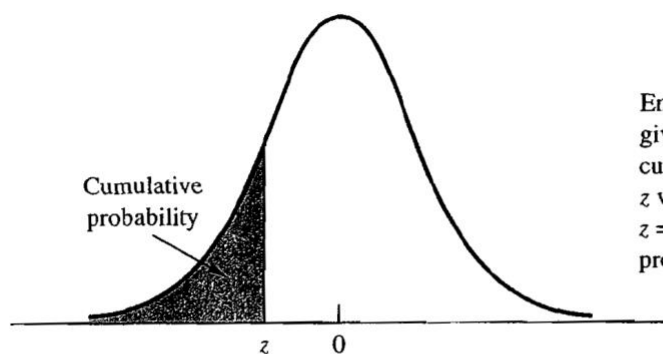
CUMULATIVE PROBABILITIES FOR THE STANDARD NORMAL DISTRIBUTION



Entries in the table give the area under the curve to the left of the z value. For example, for $z = 1.25$, the cumulative probability is .8944.

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9913
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9986	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990

CUMULATIVE PROBABILITIES FOR THE STANDARD NORMAL DISTRIBUTION



Entries in this table give the area under the curve to the left of the z value. For example, for $z = -0.85$, the cumulative probability is .1977.

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
-2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
-2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
-2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
-2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
-2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
-2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
-2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
-2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
-2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
-2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
-1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
-1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
-1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
-1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
-1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
-1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
-1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
-1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
-1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
-1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
-.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
-.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
-.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
-.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
-.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
-.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
-.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
-.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
-.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
-.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641

t Table

cum. prob	<i>t</i> _{.50}	<i>t</i> _{.75}	<i>t</i> _{.80}	<i>t</i> _{.85}	<i>t</i> _{.90}	<i>t</i> _{.95}	<i>t</i> _{.975}	<i>t</i> _{.99}	<i>t</i> _{.995}	<i>t</i> _{.999}	<i>t</i> _{.9995}
one-tail	0.50	0.25	0.20	0.15	0.10	0.05	0.025	0.01	0.005	0.001	0.0005
two-tails	1.00	0.50	0.40	0.30	0.20	0.10	0.05	0.02	0.01	0.002	0.001
df											
1	0.000	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.31	636.62
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.000	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.000	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.000	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.000	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.000	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.000	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	0.000	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	0.000	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	0.000	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	0.000	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	0.000	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	0.000	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	0.000	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	0.000	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	0.000	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	0.000	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646
40	0.000	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	0.000	0.679	0.848	1.045	1.296	1.671	2.000	2.390	2.660	3.232	3.460
80	0.000	0.678	0.846	1.043	1.292	1.664	1.990	2.374	2.639	3.195	3.416
100	0.000	0.677	0.845	1.042	1.290	1.660	1.984	2.364	2.626	3.174	3.390
1000	0.000	0.675	0.842	1.037	1.282	1.646	1.962	2.330	2.581	3.098	3.300
Z	0.000	0.674	0.842	1.036	1.282	1.645	1.960	2.326	2.576	3.090	3.291
	0%	50%	60%	70%	80%	90%	95%	98%	99%	99.8%	99.9%
	Confidence Level										

F critical values

	Area in Upper Tail											
		Numerator Degrees of Freedom										
		1	2	3	4	5	6	7	8	9	10	
Denominator Degrees of Freedom	1	0.050	161.45	199.50	215.71	224.58	230.16	233.99	236.77	238.88	240.54	241.88
		0.025	647.79	799.50	864.16	899.58	921.85	937.11	948.22	956.66	963.28	968.63
	2	0.050	18.51	19.00	19.16	19.25	19.30	19.33	19.35	19.37	19.38	19.40
		0.025	38.51	39.00	39.17	39.25	39.30	39.33	39.36	39.37	39.39	39.40
	3	0.050	10.13	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81	8.79
		0.025	17.44	16.04	15.44	15.10	14.88	14.73	14.62	14.54	14.47	14.42
	4	0.050	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.96
		0.025	12.22	10.65	9.98	9.60	9.36	9.20	9.07	8.98	8.90	8.84
	5	0.050	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74
		0.025	10.01	8.43	7.76	7.39	7.15	6.98	6.85	6.76	6.68	6.62
	6	0.050	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06
		0.025	8.81	7.26	6.60	6.23	5.99	5.82	5.70	5.60	5.52	5.46
	7	0.050	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64
		0.025	8.07	6.54	5.89	5.52	5.29	5.12	4.99	4.90	4.82	4.76
	8	0.050	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35
		0.025	7.57	6.06	5.42	5.05	4.82	4.65	4.53	4.43	4.36	4.30
	9	0.050	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14
		0.025	7.21	5.71	5.08	4.72	4.48	4.32	4.20	4.10	4.03	3.96
	10	0.050	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98
		0.025	6.94	5.46	4.83	4.47	4.24	4.07	3.95	3.85	3.78	3.72
	11	0.050	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90	2.85
		0.025	6.72	5.26	4.63	4.28	4.04	3.88	3.76	3.66	3.59	3.53
	12	0.050	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80	2.75
		0.025	6.55	5.10	4.47	4.12	3.89	3.73	3.61	3.51	3.44	3.37
	13	0.050	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71	2.67
		0.025	6.41	4.97	4.35	4.00	3.77	3.60	3.48	3.39	3.31	3.25
	14	0.050	4.60	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65	2.60
		0.025	6.30	4.86	4.24	3.89	3.66	3.50	3.38	3.29	3.21	3.15
	15	0.050	4.54	3.68	3.29	3.06	2.90	2.79	2.71	2.64	2.59	2.54
		0.025	6.20	4.77	4.15	3.80	3.58	3.41	3.29	3.20	3.12	3.06

F critical values

Denominator degrees of freedom	Area in Upper Tail	Numerator Degrees of Freedom									
		1	2	3	4	5	6	7	8	9	10
16	0.050	4.49	3.63	3.24	3.01	2.85	2.74	2.66	2.59	2.54	2.49
	0.025	6.12	4.69	4.08	3.73	3.50	3.34	3.22	3.12	3.05	2.99
17	0.050	4.45	3.59	3.20	2.96	2.81	2.70	2.61	2.55	2.49	2.45
	0.025	6.04	4.62	4.01	3.66	3.44	3.28	3.16	3.06	2.98	2.92
18	0.050	4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.51	2.46	2.41
	0.025	5.98	4.56	3.95	3.61	3.38	3.22	3.10	3.01	2.93	2.87
19	0.050	4.38	3.52	3.13	2.90	2.74	2.63	2.54	2.48	2.42	2.38
	0.025	5.92	4.51	3.90	3.56	3.33	3.17	3.05	2.96	2.88	2.82
20	0.050	4.35	3.49	3.10	2.87	2.71	2.60	2.51	2.45	2.39	2.35
	0.025	5.87	4.46	3.86	3.51	3.29	3.13	3.01	2.91	2.84	2.77
25	0.050	4.24	3.39	2.99	2.76	2.60	2.49	2.40	2.34	2.28	2.24
	0.025	5.69	4.29	3.69	3.35	3.13	2.97	2.85	2.75	2.68	2.61
30	0.050	4.17	3.32	2.92	2.69	2.53	2.42	2.33	2.27	2.21	2.16
	0.025	5.57	4.18	3.59	3.25	3.03	2.87	2.75	2.65	2.57	2.51
35	0.050	4.12	3.27	2.87	2.64	2.49	2.37	2.29	2.22	2.16	2.11
	0.025	5.48	4.11	3.52	3.18	2.96	2.80	2.68	2.58	2.50	2.44
40	0.050	4.08	3.23	2.84	2.61	2.45	2.34	2.25	2.18	2.12	2.08
	0.025	5.42	4.05	3.46	3.13	2.90	2.74	2.62	2.53	2.45	2.39
45	0.050	4.06	3.20	2.81	2.58	2.42	2.31	2.22	2.15	2.10	2.05
	0.025	5.38	4.01	3.42	3.09	2.86	2.70	2.58	2.49	2.41	2.35
50	0.050	4.03	3.18	2.79	2.56	2.40	2.29	2.20	2.13	2.07	2.03
	0.025	5.34	3.97	3.39	3.05	2.83	2.67	2.55	2.46	2.38	2.32
55	0.050	4.02	3.16	2.77	2.54	2.38	2.27	2.18	2.11	2.06	2.01
	0.025	5.31	3.95	3.36	3.03	2.81	2.65	2.53	2.43	2.36	2.29
60	0.050	4.00	3.15	2.76	2.53	2.37	2.25	2.17	2.10	2.04	1.99
	0.025	5.29	3.93	3.34	3.01	2.79	2.63	2.51	2.41	2.33	2.27
90	0.050	3.95	3.10	2.71	2.47	2.32	2.20	2.11	2.04	1.99	1.94
	0.025	5.20	3.84	3.26	2.93	2.71	2.55	2.43	2.34	2.26	2.19
92	0.050	3.94	3.10	2.70	2.47	2.31	2.20	2.11	2.04	1.98	1.94
	0.025	5.19	3.84	3.26	2.93	2.71	2.55	2.43	2.33	2.26	2.19
94	0.050	3.94	3.09	2.70	2.47	2.31	2.20	2.11	2.04	1.98	1.93
	0.025	5.19	3.84	3.26	2.93	2.70	2.55	2.43	2.33	2.25	2.19
96	0.050	3.94	3.09	2.70	2.47	2.31	2.19	2.11	2.04	1.98	1.93
	0.025	5.19	3.83	3.26	2.92	2.70	2.54	2.42	2.33	2.25	2.18
98	0.050	3.94	3.09	2.70	2.46	2.31	2.19	2.10	2.03	1.98	1.93
	0.025	5.18	3.83	3.25	2.92	2.70	2.54	2.42	2.32	2.25	2.18
96	0.050	3.94	3.09	2.70	2.47	2.31	2.19	2.11	2.04	1.98	1.93
	0.025	5.19	3.83	3.26	2.92	2.70	2.54	2.42	2.33	2.25	2.18

F critical values

Denominator Degrees of Freedom	Area in Upper Tail	Numerator Degrees of Freedom									
		15	16	17	18	19	20	25	30	35	40
5	0.050	4.62	4.60	4.59	4.58	4.57	4.56	4.52	4.50	4.48	4.46
	0.025	6.43	6.40	6.38	6.36	6.34	6.33	6.27	6.23	6.20	6.18
10	0.050	2.85	2.83	2.81	2.80	2.79	2.77	2.73	2.70	2.68	2.66
	0.025	3.52	3.50	3.47	3.45	3.44	3.42	3.35	3.31	3.28	3.26
15	0.050	2.40	2.38	2.37	2.35	2.34	2.33	2.28	2.25	2.22	2.20
	0.025	2.86	2.84	2.81	2.79	2.77	2.76	2.69	2.64	2.61	2.59
20	0.050	2.20	2.18	2.17	2.15	2.14	2.12	2.07	2.04	2.01	1.99
	0.025	2.57	2.55	2.52	2.50	2.48	2.46	2.40	2.35	2.31	2.29
25	0.050	2.09	2.07	2.05	2.04	2.02	2.01	1.96	1.92	1.89	1.87
	0.025	2.41	2.38	2.36	2.34	2.32	2.30	2.23	2.18	2.15	2.12
30	0.050	2.01	1.99	1.98	1.96	1.95	1.93	1.88	1.84	1.81	1.79
	0.025	2.31	2.28	2.26	2.23	2.21	2.20	2.12	2.07	2.04	2.01
35	0.050	1.96	1.94	1.92	1.91	1.89	1.88	1.82	1.79	1.76	1.74
	0.025	2.23	2.21	2.18	2.16	2.14	2.12	2.05	2.00	1.96	1.93
40	0.050	1.92	1.90	1.89	1.87	1.85	1.84	1.78	1.74	1.72	1.69
	0.025	2.18	2.15	2.13	2.11	2.09	2.07	1.99	1.94	1.90	1.88
45	0.050	1.89	1.87	1.86	1.84	1.82	1.81	1.75	1.71	1.68	1.66
	0.025	2.14	2.11	2.09	2.07	2.04	2.03	1.95	1.90	1.86	1.83
50	0.050	1.87	1.85	1.83	1.81	1.80	1.78	1.73	1.69	1.66	1.63
	0.025	2.11	2.08	2.06	2.03	2.01	1.99	1.92	1.87	1.83	1.80
55	0.050	1.85	1.83	1.81	1.79	1.78	1.76	1.71	1.67	1.64	1.61
	0.025	2.08	2.05	2.03	2.01	1.99	1.97	1.89	1.84	1.80	1.77
60	0.050	1.84	1.82	1.80	1.78	1.76	1.75	1.69	1.65	1.62	1.59
	0.025	2.06	2.03	2.01	1.98	1.96	1.94	1.87	1.82	1.78	1.74

Part I Solutions

1. A $n = \left(\frac{z_{\alpha/2}}{E} \right)^2 p^*(1-p^*) = \left(\frac{1.96}{0.08} \right)^2 (.15)(.85) = 76.53 \rightarrow 77$

2. D

	Default	Not Default	
Late twice	$0.95*0.03=0.0285$	$0.1*0.97=0.097$	0.1255
Not late twice	$0.05*0.03=0.0015$	$0.9*0.97=0.873$	0.8745
	0.03	0.97	1.00

$$P(\text{Default} \mid \text{Late twice}) = 0.0285/0.1255 = 0.2271$$

3. A The mean of a t-distribution is zero.

4. A Using the formula for $n = \left(\frac{z_{\alpha/2}}{E} \right)^2 p^*(1-p^*)$, first compare (i) and (ii). You will see that $n_1 \leq n_2$. Then compare (ii) and (iii). You will see that $n_2 \leq n_3$.

5. B $P(\bar{X} > 550) = P(Z > \frac{550-540}{15/\sqrt{9}}) = P(Z > 2) = 1 - .9772 = 0.0228$.

6. D $P(X > 550) = P(Z > \frac{550-540}{15}) = P(Z > .67) = 1 - .7486 = 0.2514$.

7. B $n = \left(\frac{z_{\alpha/2}\sigma}{E} \right)^2 = \left(\frac{1.645*7}{1.2} \right)^2 = 92.08 \rightarrow 93$.

8. C Recall that the burden of proof lies with H_a .

9. D Since the mean of the population is already given as 30, not rejecting $H_0: \mu \leq 34$ means correct conclusion is made. So no error is committed.

10. C Normal population, population standard deviation is unknown -- this is one case when t statistic is used to test a hypothesis.
 $p\text{-value} = 2 P(T < -1.73) = 2 P(T > 1.73) = 2 (0.05) = 0.10$. (Look at the 19 degrees of freedom row of the t-table).

11. B In a two-tailed hypothesis test, if the confidence interval does not include the hypothesized value (here, 3.5), the null hypothesis is rejected.

12. B $E(X) = 0(.40) + 1(.30) + 2(.20) + 3(.10) = 1.0$

13. C Since it's a two-tailed test, $p\text{-value} = 2 P(Z > 1.46) = 2 (0.072) = 0.144$. Since the $p\text{-value} > 0.10$, we cannot reject H_0 at $\alpha = 10\%$.

14. B $\sigma_{\bar{p}} = \text{stdev}(\bar{p}) = \sqrt{\frac{p_0(1-p_0)}{n}} = \sqrt{\frac{.88(1-.88)}{66}} = 0.04$,

$$z = \frac{\bar{p} - p_0}{\sigma_{\bar{p}}} = \frac{0.80 - 0.88}{0.04} = -2.0$$

15. B $t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{71.2 - 68}{18/\sqrt{42}} = 1.152$

16. A Since $p\text{-value} = P(t > 1.152) = 0.1280 > \alpha = 0.10$, do not reject H_0 .

17. C $\mu_0 + t_{\alpha, n-1} \frac{s}{\sqrt{n}} = 68 + 1.302 \frac{18}{\sqrt{42}} = 71.618.$
18. B Since the data set is a matched/paired sample and here we are interested to determine whether there is a significant **difference** in the MPG of the two brands of automobiles, the hypotheses are $H_0: \mu_d = 0$ and $H_a: \mu_d \neq 0.$
19. D $\bar{d} = \frac{\sum d_i}{n} = \frac{(-2+3+0+3)}{4} = 1, \quad s_d = \sqrt{\frac{\sum (d_i - \bar{d})^2}{n-1}} = \sqrt{\frac{18}{3}} = 2.45$
- $$t = \frac{\bar{d} - \mu_d}{s_d / \sqrt{n}} = \frac{1-0}{2.45 / \sqrt{4}} = 0.816$$
20. A Since $t = 0.816 < t_{(.05, 3)} = 2.353$, do not reject H_0 .
21. D Obvious.
22. C Obvious.
23. C Since “0” is not within the confidence interval $[1.75, 3.92]$, we reject $H_0: \beta_1 = 0$.
24. A $b_1 = \frac{SS_{xy}}{SS_x} = \frac{-195.50}{182.83} = -1.069, \quad b_0 = \bar{y} - b_1 \bar{x} = \frac{165}{6} - (-1.069)\left(\frac{77}{6}\right) = 41.22$
- Hence, Average Rental Rate = $41.22 - 1.069 \times \text{Vacancy Rate}$
25. B Test statistic: $F = \frac{MSR}{MSE} = 8.32$, Critical value: $F_{0.05, 1, 4} = 7.71$
26. C Since $F = 8.32 > 7.71$, reject $H_0: \beta_1 = 0$.
27. D $\hat{y} = 41.22 - 1.069(20) = 19.84$
28. B $z = 1.04$, Minimum Value is $x = 30000 + 1.04(8000) = 38,320$
29. D $P(X < 19760) = P\left(\frac{X - 30000}{8000} < \frac{19760 - 30000}{8000}\right) = P(Z < -1.28) = 0.10$
30. A $P(X < 3) = P(X \leq 2) = (.2059 + .3432 + .2669) = 0.816$
31. C Look at the properties R^2 and R_a^2 in the lecture note on multiple regression.
32. D As we add variables to the model, R^2 does not decrease
33. C Adjusted $R^2 = 1 - \frac{12.45}{537.48/18}$
34. D $VIF = \frac{1}{1 - 0.225}$
35. A For average satisfaction, use the confidence interval

Part II Solutions

Problem 1.

a. 7

- b. $H_0: \mu_1 - \mu_2 = 0$
 $H_a: \mu_1 - \mu_2 \neq 0$

$$c. \quad t = \frac{\bar{x}_1 - \bar{x}_2 - 0}{s_{\bar{x}_1 - \bar{x}_2}} = 2.60$$

a. $0 < 0.99 < \mu_1 - \mu_2 < 13.01$, $\mu_2 > \mu_1$ with 95% confidence.

e. Since 0 is not within the confidence interval (0.99, 13.01), reject H_0 with $\alpha = 5\%$. We would conclude that there is a significant difference between the mean project completion times for the current technology and the new software package.

Problem 2.

a. The relationship is significant since p-value = 0.000 (t or F test)

b. $\hat{y}_8 = 6.090 + 0.729 \times 8 \approx 11.922$. $y_8 - \hat{y}_8 = 11 - 11.922 = -0.922$

c. Look at the MINITAB output for 95% PI for 14 years of experience (4th observation). (11.244, 21.348)

d. Yes. Check p-value.

e. The confidence interval is

$$\left(b_1 - t_{\alpha/2, n-2} s_{b_1}, b_1 + t_{\alpha/2, n-2} s_{b_1} \right) = (0.72897 - 2.228 \times 0.0947, 0.72897 + 2.228 \times 0.0947) = (0.5180, 0.9400).$$

The above confidence interval does not contain 0.45. We reject the Null hypothesis that this coefficient is 0.45 with 5% level of significance

Problem 3.

a. $H_0: \beta_1 = \beta_2 = \beta_3 = \beta_4 = \beta_5 = 0$

H_A : at least one of $\beta_1, \beta_2, \beta_3, \beta_4$, and β_5 is not equal to 0

Decision rule: if $F = \frac{MSR}{MSE} > F_{5,87,5\%}$, we reject H_0 at a 5% significance level and conclude at least one of $\beta_1, \beta_2, \beta_3, \beta_4$, and β_5 is not equal to 0.

Conclusion: Since $F = 18.4 > F_{5,87,5\%}$ (between 2.31 and 2.37), we reject H_0 .

Alternative method: If the p-value $< 5\%$, we reject H_0 . Since the p-value = .000, we reject H_0 .

b. $H_0: \beta_4 = \beta_5 = 0$

H_A : at least one of β_4 and β_5 is not zero

Decision rule: if $F = \frac{\frac{SSR(\text{Full}) - SSR(\text{Reduced})}{2}}{MSE(\text{Full})} > F_{2,87,5\%}$, we reject H_0 at a 5% significance level.

$$F = \frac{\frac{4330337692 - 2507213192}{2}}{47062256} = 19.369$$

Conclusion: Since $F = 19.369 > F_{2,87,5\%}$ (between 3.09 and 3.15), we reject H_0 and conclude that a significant difference existed between male and female workers.

c. For males,

$$\begin{aligned}\text{The predicted salary} &= 50391 + 994(12) + 18.5(120) + 298(15) - 3174(1) + 978(12)(1) \\ &= 77571\end{aligned}$$

For females,

$$\begin{aligned}\text{The predicted salary} &= 50391 + 994(12) + 18.5(120) + 298(15) - 3174(0) + 978(12)(0) \\ &= 69009\end{aligned}$$